LYRA-CPP — THETIS DIRECT-PORT PLAN (TX)

Authoritative doc for the current TX work direction.

> *[!] The older* `EXECUTION_PLAN.md` *(Step 14 wire-layer rebuild, Phase 3 etc.) describes a DIFFERENT approach that is now superseded for the TX work. Do NOT mix the two plans — that doc tracks an earlier wire-rebuild pass which was partial and Lyra-native-leaning. THIS doc is the active plan going forward. Old doc remains in tree for historical reference only (some shipped pieces from it are still in production — noted per-component in the registry below).*

Branch: `tx-rebuild` (the direct-port work lives here; `origin/main` carries the earlier Step 14 / TX-1 line and is deliberately untouched by the direct-port arc).

Reference tree (the ONLY trusted TX reference):

- `D:\sdrprojects\OpenHPSDR-Thetis-2.10.3.13\Project Files\Source\`
  - `wdsp\*.c` — WDSP DSP engine (GPL v3+, port-with-attribution OK)
  - `ChannelMaster\*.c` — wire layer (GPL v3+, port-with-attribution OK)
  - `Console\*.cs` — C# UI/control glue (study ONLY; write Lyra-native equivalents)
- `Y:\Claude local\_hl2src\` — HL2+ ak4951v4 gateware RTL

(`control.v` / `radio.v` / `usopenhpsdr1.v` etc. — the ground truth when host-side reference and gateware disagree)

Provenance & attribution rule (operator-locked 2026-06-09, supersedes the 2026-05-31 "no-attribution-in-commits" tightening):

Lyra-cpp is a derivative work of openHPSDR Thetis (ChannelMaster + WDSP modules), GPL v3+. The GitHub project description openly states this: *"TX baseline ported from openHPSDR Thetis (ChannelMaster) with Lyra-native DSP additions."* Open attribution is both legally required (GPL) and operator-intended (public project posture).

1. **License attribution at file head — REQUIRED.** Any file ported from WDSP or ChannelMaster carries a header block crediting upstream (NR0V for WDSP modules; openHPSDR Thetis for ChannelMaster modules) + the GPL v3+ notice. Existing `AAMix.{h,cpp}`, `CMaster.cpp`, `RadioNet.cpp` etc. already follow this; keep it.
2. **Code comments citing source — ENCOURAGED.** Citations may name the reference openly (`Reference: Thetis ChannelMaster cmaster.c:297-313`, `// Per openHPSDR networkproto1.c:1078`, `// HL2+ ak4951v4 gateware control.v:209-220`). The file:line is the primary technical content; the reference-app name is context that helps future readers find the source tree.
3. **Commit messages — REFERENCE-APP NAMES PERMITTED.** A commit like *"port aamix.c from Thetis ChannelMaster with Lyra-native idiom translations"* is fine — it matches the public GitHub posture and makes the history grep-able. The earlier `4cef88d` tightening (2026-05-31) that put commit messages on the no-attribution list is **rescinded**.
4. **Public docs (README, `docs/*.md` design docs, memory files, this plan) — OPEN ATTRIBUTION.** Same posture as the GitHub description.
5. **User-facing UI strings — KEEP OPERATOR-FOCUSED.** UI labels operators see in normal operation (panel titles, button text, error toasts) describe Lyra behavior, not provenance. *"AAMix output (HL2 jack)"* not *"Thetis-ported AAMix output"*. About dialog / credits screen / install docs MAY (and probably should) name openHPSDR Thetis + WDSP/NR0V openly — credit where due. This is the only category that stays "Lyra-native voice in operator-facing labels," because operators interact with Lyra, not its source provenance.
6. **The "Lyra-Native style governs surrounding architecture only" rule (Stage B-locked) STANDS unchanged.** Reference DSP API call patterns are byte-for-byte faithful; Qt/Vulkan/QML/process model is Lyra-native.
7. **Reference precedence STANDS unchanged.** Verify TX ONLY against Thetis 2.10.3.13 (`D:\sdrprojects\OpenHPSDR-Thetis-2.10.3.13\...`) + the HL2+ ak4951v4 gateware RTL (`Y:\Claude local\_hl2src\`). **Old**

Python Lyra is NOT a trusted TX reference (its TX never worked).

Consequence of the 2026-06-09 amendment: the three historical commits (`efdde02`, `604d87b`, `ed9f0cd`) that "predated the no-attribution rule" and contained "Thetis" in their messages are **no longer in violation** — they're now consistent with the rule. Task #73 ("Cleanup pre-existing 'Thetis' mentions in shipped code") is **OBSOLETED** by this amendment.

LOCKED METHODOLOGY (Stage B-validated, do NOT relax)

These rules were earned through the Stage B aamix.c port arc (2026-06-08) — see SESSION_LOG.md for the full trail.

1. **Smallest revertable empirical step -> operator HL2 bench ->

next step.** Each commit is one focused change with a falsifiable bench gate. Multi-step "design then implement" is for stages that genuinely need it (operator approves); default is iterate-and-bench.

2. **"Do as Thetis does, Lyra-Native style"** (locked

2026-05-30 after the `fexchange0` crash arc, re-validated tonight after the B.6.b shortcut attempt cost a revert cycle). Three sub-rules:

- Every WDSP API call matches the reference byte-for-byte

(parameter values, order, the SET of setters fired, the lifecycle stage at which they fire). If tempted to pick a value the reference doesn't use, STOP and find what the reference uses. "I have a reason to pick differently" means I don't. Reference has been right every time.

- "Lyra-Native style" governs surrounding architecture ONLY

(Qt + Vulkan + QML + process/thread model + facade APIs + IPC + UI shell). The WDSP DSP-engine call surface is reference-bound.

- Provenance only in code COMMENTS + design docs + memory.

Never commit messages, never UI strings.

3. **Counters first when the symptom is "nothing happens".**

The Stage B 2-stage instrumentation (chain counters in xMixAudio/mix_main + dispatch counters in dispatchAudioFrame) localized the silent-audio defect to a single line of code on the 2nd bench. Pure speculation would have cost a multi-round revert/speculate cycle. Atomic counter + rate-limited qInfo = trivial overhead, very high diagnostic ROI when "I press play, hear nothing, log looks normal."

4. **Operator-empirical rule overrides agent inference.**

When operator hardware data contradicts what an audit/red-team/Plan-agent concluded, the data wins. Do not defend the audit — re-open, bisect, fix. This rule has fired repeatedly through the broader project arc; the Stage B B.6.b-fix1 fix was an instance (chain counters healthy suggested AAMix was correct; the temptation was to revert the whole port; the empirical pattern instead pointed at the wire-up surface where the bug actually was).

5. **Read the reference FIRST, in shipped-code form, before

designing.** Every time we designed first and read second, we shipped a regression. Every time we read first, we got the right answer in one pass. The no-attribution rule applies to shipped code; it does NOT mean "don't read the reference." Read it, dossier the file:line in the plan doc, then write Lyra-cpp.

6. **Session-start discipline:** `git fetch origin` is the

FIRST action every session, before reading any local state. If divergence exists, surface it to the operator BEFORE writing code. No exceptions.

7. **Push after every component ships, not at EOD.** Each

clean operator HL2 bench -> backup bundle + push immediately. Smaller in-flight delta = smaller surprise blast radius.

8. **Backup bundle naming:** `_backups/lyra-cpp-YYYY-MM-DD-.bundle`.

`git bundle create … --all` (covers all refs).

STATUS REGISTRY — what's done, in-progress, pending

■ ***2026-06-13 — TX BRING-UP COMPLETE (P0 → P4.b) + SHIPPED.***

*The verbatim direct-port wire chain is LIVE and operator-HL2-bench-passed: SSB voice both sidebands, Phase-3-EXIT RF-safety kill-test, TUN Thetis-exact zero-beat, and **MSHV → TCI → FT8 real on-air QSOs** (incl. F4FLF ~4160 mi DX). §7 dead-code retirements done. Released as **v0.2.3** (*`main` == `tx-rebuild` == `e8aafbc`*; tags* `v0.2.3` + `tx-working-2026-06-13`*; GitHub Release + installer published). Normal feature dev resumes off* `main`*. Next milestones: PC/VAC mic-input paths (#102/#103/#104), then v0.3 PureSignal (the P0.d* `_cmaster` *struct already carries the full PS surface — no retrofit).*

■ ***2026-06-15 — VAC (#102/#158) SHIPPED as v0.3.0; post-P4 ordering LOCKED.*** *The next-milestone PC/VAC mic-input path shipped: the* `ivac.c` → `wire/IVAC` *device-layer rework (single full-duplex PortAudio stream, VAC/TCI crash fixes, RX-muted-on-TX) released as **v0.3.0** (*`main` == `tx-rebuild`*). **Open loose-end from that arc:** the TX MIC/ALC meters were re-homed onto* `GetTXAMeter(chid(1,0)=1)` **inside* the VAC work ("#158 (post-DL) re-home" in* `metermodel.cpp` / `wdsp_engine.cpp`*) but never got their own bench gate — operator reports MIC/ALC read nothing. Treat as **UNVERIFIED, not done** (the WDSP-TX-chain leveler/ALC rows are* `[WIP] "confirm reference-parity"`*). Closing it = a deep Thetis read of the TX-meter read/display path (which* `txaMeterType`*, which channel, where in the chain) + port + bench gate. Code-level contributors found so far (candidates, NOT yet root-caused — do not patch blind): the secondary-readout line is still hard-stubbed* `"—"` *(*`formatSecondaryText`*, metermodel.cpp:577-582); and the AK4951 codec mic delivers zero into the TXA until a rate change because the at-open seed (hl2_stream.cpp:837) sets the rate bits but not* `mic_decimation_factor` *(defaults 0 = the* `++count == factor` *harvest gate never fires).*

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LOCKED post-P4 feature ordering (operator decision 2026-06-15):

*1. **TX MIC/ALC metering** — Thetis-reference read + port + bench (closes out the VAC arc; verify-first, counters-first per the locked methodology — do NOT patch the candidates above blind).*

*2. **Stage F — TX DSP chain** (EQ / compressor-Combinator / leveler / ALC + their meters; broad TX value; feeds the in-flight #49 Profile Manager).*

*3. **RX2 + Split** (#96–#101) — daily-use value, AND it builds the* `(mox, ps_armed, rx2_enabled)` *DDC-dispatch state machine that PS reuses (RX2 lives on DDC1; PS commandeers DDC0/DDC1 during MOX+PS).*

*4. **PureSignal (Stage G/H, v0.3)** — LAST. Narrowest audience (needs* *the HL2 hardware mod + PS gateware), deepest / most cross-cutting surface; lands on top of the dispatch foundation RX2 builds, no wire retrofit (the P0.d `_cmaster` struct already carries the PS surface). Rationale: PS gets cheaper + safer AFTER RX2, not merely lower-priority. The plan's F-before-G was version cadence (v0.2.x dynamics → v0.3 PS), NOT a hard compile dependency — so RX2 slots between F and PS.*

■ ***2026-06-17 — VAC closeout v0.3.1 + Stage-F native TX DSP rack + Profiles + TX monitor SHIPPED (post-P4).** All native (Lyra-side, NOT direct ports — see the SHIPPED-WORK LOG entry for hashes).* ***v0.3.1*** *(`a3a517f`, 2026-06-15) closed the VAC arc: the enable/disable crash fix, Vol/Mute routed to VAC, AND the TX MIC/ALC/LEV metering the v0.3.0 banner above flagged UNVERIFIED is now wired Thetis-format (`9594bd5` #160) — so that loose-end is* ***resolved in code*** *(the WDSP leveler/ALC rows below are flipped to [DONE] for their cffi bindings + metering wiring). NOTE: the #160 metering **bench close-out + Brent/Timmy v0.3.1 field-confirmation are still IN PROGRESS** — the meters are shipped, not yet field-validated.*

Stage F TX DSP rack (native):

8-band parametric EQ (#50), Speech processor / Auto-AGC leveller + de-esser (#88), X-Air-style Combinator (#51), Plate (#52), with digital-mode auto-bypass + digital/CW lamp dim (#163). ***Profiles #49*** *rack capture/apply (schema v3) + sideband/band-agnostic SSB profile + panel-lock fix + Speech-profile capture/lamp refresh (#162).* ***TX audio monitor #90*** *— 3 reference-faithful routes (HL2 jack / VAC device / TCI RX-audio) + the inline Audio-panel "Out" picker & MON→MON-TX relabel (#164). Released as* ***v0.3.1*** *(`main` == `tx-rebuild`); the native DSP/profile/monitor work rides on top (HEAD `dc45485`).* ***NEXT (current open queue — one task):*** *an AM/DSB RX-DSP bug — AM and DSB produce only the upper sideband (lower missing); the WDSP passband for AM/DSB is likely set one-sided instead of symmetric -W..+W. (GitHub / main-branch posture questions pending operator discussion.)*

Format: per-file or per-component row. Status tags (ASCII so they render identically in the PDF/DOCX siblings):

• [DONE] DONE + operator-bench-validated

• [WIP] IN-PROGRESS (committed but not yet bench-validated, or partially shipped)

• [PEND] PENDING (planned, not started)

• [N/A] N/A (Lyra-native equivalent already shipped; no direct port planned — see notes column)

Wire layer (ChannelMaster — port directly w/ attribution)

Source	Target (lyra-cpp)	Status	Notes
WDSP exports linkage (`OpenChannel`/`fexchange0`/analyzers/PS family)	`src/wire/wdspcalls.{h,cpp}` (P0.a)	[DONE]	**P0.a** `e0c0f4a` **2026-06-09.** The single operator-approved linkage seam: fn-ptr table carrying the WDSP exports' exact names, resolved once from the loaded wdsp.dll. PureSignal entry family complete (verified vs dumpbin). RESAMPLE-typed + `flush_resample` added at P0.c.
`ChannelMaster\cmcomm.h` (family common header)	`src/wire/cmcomm.{h,cpp}` (P0.b -> P0.d)	[DONE]	`complex` typedef + PORT + verbatim malloc0 (wdsp/utilities.c:37-43) + PI + the reference include surface. P0.d added the opaque verbatim twin typedefs for deferred-subsystem types (ANB/NOB/VOX/TXGAIN/ANALYZERS + DELAY at P2.a); **EER completed at P2.a 2026-06-12** — full verbatim struct (wdsp/eer.h:30-49) + the umbrella-mapping note (.cpp files include explicit family headers — an include-list umbrella would be circular under `#pragma once`).
`ChannelMaster\ilv.c` (TX I/Q interleaver)	`src/wire/ILV.{h,cpp}` (P0.b)	[DONE]	**P0.b** `1f49d98` **2026-06-11.** VERBATIM direct port (`ilv`, `*ILV` twin typedef, raw Outbound fn ptr, `pcm->xmtr[].pilv` setters, malloc0/_aligned_free). Mechanical diff vs ilv.{h,c} = whitespace-only. The earlier `src/wdsp/ILV` retrofit + its pilv[] bank are retired. scratch/test_ilv ALL PASS.
`ChannelMaster\cmbuffs.{h,c}` (CMB ring + cm_main pump)	`src/wire/CmBuffs.{h,cpp}` (P0.d)	[DONE]	**P0.d** `afc7950` **2026-06-12.** VERBATIM rewrite (`cmb`, `*CMB`, `#define CMB_MULT (3)`, malloc0/_aligned_free; the Phase-C retrofit's calloc/intptr_t/guard deviations removed; `pcm->in[]` alloc moved back to create_cmaster per reference). Mechanical diff = whitespace-only. [DONE] **P0.d operator HL2 RX-regression bench PASSED 2026-06-12.**
`ChannelMaster\cmsetup.{h,c}` (radio structure + id helpers)	`src/wire/cmsetup.{h,cpp}` (P0.d, NEW)	[DONE]	**P0.d** `afc7950`**.** VERBATIM: cmMAX* sizing macros (16/4/4/2/32), rxid/txid/sp0id/stype/chid/inid/mixinid/getbuffsize, SetRadioStructure + set_cmdefault_rates. main.cpp config: SetRadioStructure(2,1,1,1,0,…) -> chid(0,0)=0 RX1 / chid(1,0)=1 TX matches the live channel layout. [DONE] Bench PASSED 2026-06-12.

Source	Target (lyra-cpp)	Status	Notes
`ChannelMaster\cmaster.{h,c}` (FULL: struct + create/destroy + xcmaster + setters)	`src/wire/CMaster.{h,cpp}` (P0.d)	[DONE]	**P0.d** `afc7950`**.** VERBATIM rewrite: full `_cmaster` struct (cmaster.h:39-99 incl. the PS surface — out[3]/panalalloc/pgain/peer), `cmaster,*CMASTER`, raw TCI callback fn ptrs, `enum AudioCODEC`, `cm = {0}`. Verbatim no-arg create_xmtr/destroy_xmtr (out[0..2] malloc0 + OpenChannel(ch 1) + XCreateAnalyzer(TX disp 1) + create_ilv through the wdspcalls seam — **the TxChannel RAII carve-out is DELETED**, `src/wdsp/TxChannel.{h,cpp}` retired). create_cmaster/destroy_cmaster verbatim per-stream loops (update[] CS + cmbuffs + in[] for all cmSTREAM streams). xcmaster verbatim (update[] CS restored, real stype/txid/chid, TCI memset restored; fexchange0 + monitor xMixAudio + xilv live). All Sendp*/Set* setters ported. Deferred subsystem lines (dexp/anti-vox/txgain/sidetone/pipe/cmasio/analyzers.c/create_rcvr-body) carried IN PLACE as reference text with DEFERRED tags; **eer RESTORED VERBATIM at P2.a 2026-06-12** (create_eer run=0 / destroy_eer / xeer / pSetEER* via the wdspcalls seam — the object exists so the sendOutbound/sendProtocol1Samples `peer->run` derefs are valid, per cmaster.c:212-224). Sole code accommodation: `(char*)""` at XCreateAnalyzer. [DONE] **Operator HL2 RX-regression bench PASSED 2026-06-12** (RX working on the new per-stream pump/ring layout).
`ChannelMaster\obbuffs.c` (TX-out ring/seam)	`src/wire/ObBuffs.{h,cpp}` (P1, NEW)	[DONE]	**P1** `75754c2` **2026-06-12** — obbuffs.{h,c} verbatim (obb/*OBB twin typedef, file-scope obp four-alias bank, the reference's own calloc/free 2014-era idioms kept). Mechanical diff: .cpp IDENTICAL; .h sole delta = the include-guard define replaced by #pragma once (documented packaging). The `sendOutbound(id, a->out)` ob_main hand-off was restored at P2.c (no longer deferred). **Wire-LIVE since P4.b 2026-06-13** — create_obbuffs/ob_main are live in the direct-port TX chain (TU no longer dormant); was [WIP] SHIPPED while dormant, now DONE in the live chain.
`ChannelMaster\networkproto1.c` (HL2 wire writer)	`src/wire/NetworkProto1.cpp` (sendProtocol1Samples) + FrameComposer write_main_loop_hl2 (LIVE)	[DONE]	**Wire-LIVE at P4.b 2026-06-13.** sendProtocol1Samples verbatim drives the EP2 writer thread; FrameComposer monolithic write_main_loop_hl2 is the WriteMainLoop_HL2 equivalent (#121/#122 fold). The old `hl2_stream.cpp` Step-14 mix retired at §7.
`ChannelMaster\network.c` (UDP + keepalive)	`src/wire/ObBuffs.{h,cpp}` ob_main + sendOutbound (LIVE)	[DONE]	**Wire-LIVE at P4.b.** sendOutbound verbatim (P2.c); the Step-14 Lyra-native OutboundRing + Ep2SendThread translations RETIRED at §7 (`fe6a438`).

Source	Target (lyra-cpp)	Status	Notes
`ChannelMaster\netInterface.c::create_rnet`	`src/wire/RadioNet.cpp::create_rnet` (LIVE)	[DONE]	Direct port shipped earlier. The Stage-A SendpOutboundRx no-op stub at line 272 was DELETED 2026-06-08 (B.6.b-fix1, commit `8b8e0da`) because it clobbered the AAMix wire-up; future codec-id switching must NOT re-introduce a default registration here. P3 = the outbound registrations (SendpOutboundTx(OutBound) etc.) — MUST register AFTER create_xmtr (the verbatim setters have NO null guards, reference ordering).

WDSP RX chain (port directly w/ attribution)

Source	Target (lyra-cpp)	Status	Notes
`ChannelMaster\aamix.{c,h}`	`src/wire/AAMix.{h,cpp}` (P0.c)	[DONE]	**P0.c** `e0f2584` **2026-06-11 — VERBATIM re-port** (supersedes the 2026-06-08 Stage-B C++23-idiom translation, which the operator rejected as a deviation class). `aamix`, `*AAMIX` twin typedef + anonymous slew struct, raw Outbound fn ptr, paamix[] bank private to the .cpp, Win32 primitives direct. Mechanical diff vs aamix.{h,c} = whitespace-only. `src/wire/resample.h` = the public RESAMPLE ABI verbatim. WdspEngine RX wire-up: capturing lambda -> static member fn + TU-scope context ptr (the reference free-fn+global shape). [DONE] **Operator HL2 RX bench PASSED 2026-06-11.**
`wdsp\firmin.c` (filter cores)	(consumed via WDSP cffi)	[N/A]	Used internally by WDSP RXA chain via cdef + dlsym. No source-level port needed (the DLL ships with Lyra-cpp via `_native/` per the Python-project pattern; cdef + dlsym in `wdsp_native.cpp`). WISDOM cache fix shipped on Python side; lyra-cpp's RX chain inherits the same path.
`wdsp\wisdom.c`	(consumed via WDSP cffi)	[N/A]	Same as firmin.c.

WDSP TX chain (per CLAUDE.md §4.1 port table)

Source	Target (lyra-cpp)	Status	Notes
`wdsp\TXA.c` (channel scaffold)	(consumed via `wire/wdspcalls` seam — P0.a/P0.d)	[DONE]	**TxChannel class DELETED at P0.d** `afc7950` — the verbatim create_xmtr in `wire/CMaster.cpp` opens the WDSP TXA channel itself (OpenChannel/fexchange0/CloseChannel through the wdspcalls seam, cmaster.c:177-190 byte-exact), which removed the runtime-DLL RAII justification the class existed for. `wdsp_native.{h,cpp}` still carries the SetTXA*/GetTXAMeter cdefs for the operator-control surface.
`wdsp\bandpass.c` (bp0/bp1)	(consumed via WDSP cffi — SetTXABandpassFreqs)	[N/A]	The §15.23 trap (SetTXABandpassRun overrides bp1 to a stale state) is documented in CLAUDE.md + project_lyra_cpp_tx.md. cffi-only access is the correct posture.
`wdsp\compress.c` (TX compressor)	`src/wdsp/Compress.{h,cpp}` OR cffi binding	[N/A]	**Superseded by the native TX DSP rack (post-P4).** Operator chose the Lyra-native path: the Speech processor (#88, `902b814`/`93adb39`) + Combinator (#51) cover speech dynamics natively, pre-WDSP-TXA. No WDSP compress.c port/cffi binding shipped.
`wdsp\cfcomp.c` (5-band CFC)	`src/wdsp/CFComp.{h,cpp}` OR cffi binding	[N/A]	**Superseded by the native Combinator #51** (`013bf12`/`61e16c4`/`b53ae35`) per the §15.19 plan — Lyra-native multiband compressor replaces the WDSP CFC. No CFC port shipped.
`wdsp\osctrl.c` (CESSB)	cffi binding (SetTXAosctrlRun)	[PEND]	Likely cffi-only; ~200 LOC port not justified unless operator UI need surfaces.
`wdsp\wcpagc.c` mode 5 (leveler)	cffi binding (SetTXALeveler*)	[DONE]	cffi binding live in the wire-LIVE TXA chain. **TX LEV metering wired Thetis-format at #160** `9594bd5` **(v0.3.1)** + #49 profile Leveler fields — addresses the v0.3.0 "UNVERIFIED" metering loose-end (shipped in code; #160 bench close-out / tester confirm still IN PROGRESS).
`wdsp\wcpagc.c` mode 5 (ALC)	cffi binding (SetTXAALCMaxGain)	[DONE]	cffi present; SetTXAALCThresh does NOT exist (§15.23-class trap — only MaxGain governs ALC ceiling). **TX ALC metering wired Thetis-format at #160** `9594bd5` **(v0.3.1)** — addresses the v0.3.0 "UNVERIFIED" flag (shipped in code; #160 bench close-out / tester confirm still IN PROGRESS).

Source	Target (lyra-cpp)	Status	Notes
`wdsp\eqp.c` (parametric EQ)	`src/wdsp/EQ.{h,cpp}` OR cffi binding	[N/A]	**Superseded by the native EQ #50** (`edbaa52/17a28b7/eb92231`) — native RBJ-biquad 8-band parametric EQ (ParamEq engine + EESDR3-style panel), wired pre-WDSP-TXA. No WDSP eqp.c port shipped.
`wdsp\gen.c` (gen0/gen1 generators)	cffi binding (SetTXAPreGen*/SetTXAPostGen*)	[WIP]	gen1 (postgen) live for TUN tone — wire-LIVE + bench-passed since P4.b 2026-06-13 (Thetis-exact zero-beat). gen0 (pregen) cdefs still deferred per §15.23 (bench-tooling nicety, not on the signal path) — hence row stays [WIP].

TX PureSignal (per CLAUDE.md §4.1 v0.3)

Source	Target (lyra-cpp)	Status	Notes
`wdsp\iqc.{c,h}`	`src/wdsp/Iqc.{h,cpp}` (~315 LOC)	[PEND]	v0.3. cffi math + Lyra-cpp wrapper for the 5-state PS lifecycle (RUN, BEGIN, SWAP, END, DONE).
`wdsp\calcc.{c,h}`	`src/wdsp/Calcc.{h,cpp}` (~1164 LOC)	[PEND]	v0.3. cffi math + Lyra-cpp wrapper for the 8-state PS FSM + 3-state auto-attenuator FSM + PSDialog UI.
`wdsp\lmath.c::xbuilder`	`src/wdsp/PsXBuilder.{h,cpp}` (~200 LOC)	[PEND]	v0.3. Cubic-spline coefficient builder, used by Lyra-cpp-side calcc orchestration.
`wdsp\delay.c`	`src/wdsp/DelayLine.{h,cpp}` (~80 LOC)	[PEND]	v0.3. TX/feedback time-alignment.

Console (C# — study only, Lyra-native equivalents)

Source	Target (lyra-cpp)	Status	Notes
`Console\console.cs::chkMOX_CheckedChanged2`	`src/ptt.{h,cpp}` + `src/wdsp_engine.cpp` keydown/keyup	[DONE]	Lyra-native FSM. Verified against the reference's keydown/keyup ordering invariants (RX-DSP stop on keydown -> wire MOX -> TX-DSP start; keyup TX-DSP off -> mox_delay -> clear MOX -> ptt_out_delay -> RX-DSP restart). **Live + operator-HL2-bench-passed at P4.b 2026-06-13** (SSB voice both sidebands + RF-safety kill-test exercise the FSM end-to-end). No direct port — C# is study-only by license.
`Console\console.cs` (TR delays)	`src/tx/TrSequencing.{h,cpp}` (LIVE)	[DONE]	Lyra-native shipped with operator's Thetis-DB-verified defaults (mox=15/rf=50/space_mox=13/ptt_out=5/key_up=10 ms).

Source	Target (lyra-cpp)	Status	Notes
`Console\console.cs::m_bATTonTX`	`src/hl2_stream.cpp` `fsmAdvance` + `compose_case_11` (LIVE wire)	[WIP]	Lyra-native. Reference posture: keydown force step-att 31 (PS-A off OR CW); keyup restore. **WIRE path traced live + functional 2026-06-17:** `fsmAdvance` raises `setTxStepAttnDb(31)` on keydown (`hl2_stream.cpp:1651`) → `tx_step_attn = 31−31 = 0` → XmitBit-gated `compose_case_11` C4 = `0x40` → ak4951v4 cmd_addr 0x0a rx_gain = 0 = min LNA during TX. **BUT NOT [DONE]:** (a) the `src/tx/AttOnTxPolicy.{h,cpp}` class is **unused / wire-inert** (its own header says so) — the live mechanism is inline in hl2_stream.cpp, the earlier doc citation was wrong; (b) ~~operator-reported 2026-06-17: NO UI surface~~ → **UI surface BUILT 2026-06-17 (§15.31):** TxPanel "ATT" lamp in the Mic↔Tune gap (gray `ATT off` / orange `ATT 31` armed-RX / solid-red `ATT -31` engaged-TX, AUTO/TUN/MOX idiom) + Settings → TX → "ATT on TX (RX-ADC protection)" group (Enable + dB 0..31 spin); operator-gated, default ON/31, QSettings `tx/attOnTx*`; `fsmAdvance` now gates on the toggle + uses the value (replaces the hardcoded `kAttOnTxDb`). Operator-approved in review; **awaiting on-air bench confirm to flip [DONE].** The `AttOnTxPolicy` class stays unused (flag: delete-or-wire-up later). Broader **Task #114 still PENDING** (panadapter TX offset + PA-enable safety).
`Console\PSForm.cs`	`src/ui/PsDialog.{h,cpp}` (NEW, v0.3)	[PEND]	v0.3 PureSignal UI. C# study only; write Qt/QML native.
`Console\HPSDR\IoBoardHl2.cs`	`src/hl2_stream.cpp` (LIVE)	[DONE]	HL2 I/O quirks implemented Lyra-native via the protocol-byte verification + gateware-RTL ground-truth pass (CLAUDE.md §15.26). Live + bench-passed across the RX + P4.b TX chain (incl. on-air FT8).

STAGE INDEX — ordered work plan

P0 verbatim-rewrite arc (2026-06-09 -> current; the ACTIVE plan)

After the operator rejected the 2026-06-09 "rule-#8" retrofit deviations ("I said PORT!"), every TX-ported file is being rewritten VERBATIM (byte-faithful; only documented packaging differences — `lyra::wire` namespace, PORT-expands-empty, the wdspcalls seam for the statically-linked WDSP exports). Each step ships with a mechanical diff vs the reference + an operator HL2 bench gate.

Step	Status	What	Commit / gate
P0.a	[DONE] DONE	`wire/wdspcalls.{h,cpp}` — the single approved WDSP linkage seam (exact export names, PS entry family complete)	`e0c0f4a` 2026-06-09
P0.b	[DONE] DONE	ilv.{h,c} verbatim -> `wire/ILV.{h,cpp}` + new `wire/cmcomm.{h,cpp}`	`1f49d98` 2026-06-11; test_ilv ALL PASS

Step	Status	What	Commit / gate
P0.c	[DONE] DONE	aamix.{h,c} verbatim -> `wire/AAMix.{h,cpp}` + `wire/resample.h` RESAMPLE ABI	`e0f2584`; [DONE] operator HL2 RX bench PASSED 2026-06-11
P0.d	[DONE] DONE	cmbuffs/cmaster/cmsetup verbatim -> `wire/CmBuffs`/`wire/CMaster`/`wire/cmsetup`; full `_cmaster` struct (PS surface); TxChannel carve-out DELETED	`afc7950` 2026-06-12; [DONE] **operator HL2 RX-regression bench PASSED 2026-06-12**
P1	[DONE]	obbuffs.c verbatim -> `wire/ObBuffs.{h,cpp}` (the TX-out seam; separate TU from cmbuffs; sendOutbound call restored at P2.c)	`75754c2` 2026-06-12; shipped dormant, then **wire-LIVE at P4.b 2026-06-13** (create_obbuffs/ob_main live in the direct-port TX chain); clean build 0 warnings; mechanical diff IDENTICAL
P2	[DONE]	sendOutbound / sendProtocol1Samples fidelity audit of the dormant wire layer — **COMPLETE 2026-06-12** (P2.a eer completion + P2.b prn outbound surface verbatim + P2.c wire/Network.cpp sendOutbound verbatim, mechanical diff IDENTICAL; the ObBuffs ob_main hand-off restored). **P2.a SHIPPED 2026-06-12** — eer completion (verbatim struct + seam entries + the four CMaster restore sites); kills the null-`peer` landmine both reference functions deref. NEXT: P2.b prn outbound surface verbatim (outLRbufp/outIQbufp/OutBufp -> reference pointer fields + the HANDLE semaphore quartet; callers bend) -> P2.c wire/Network.cpp sendOutbound verbatim (P1 branch live, ETH branch DEFERRED) + restore the ObBuffs ob_main call.	Audit finding: Lyra's Step-14 OutboundRing/Ep2SendThread are functionally-parallel idiom translations (vector buffers, bool+cv semaphores) predating the verbatim mandate — retired at P4 when the verbatim chain goes live.
P3	[DONE]	netInterface registrations + obbuffs ring lifecycle. **SHIPPED + operator HL2 bench PASSED 2026-06-12** (RX fine; stop/start + shutdown clean) — create_rnet moved to once-per-process at the main.cpp QTimer AFTER create_xmtr (the reference C#-init order; fixes the per-open prn re-allocation leak); SendpOutboundTx(OutBound) restored at the reference site (netInterface.c:1761, tail of create_rnet); UpdateRadioProtocolSampleSize verbatim (netInterface.c:1836-1858, diff IDENTICAL) called per-open per StartAudio:45 — creates obbuffs rings 0/1 (2 new ob_main pump threads per session, idle: no producers until P4); destroy_obbuffs(0/1) at close() per StopAudio:112-113. RX side deliberately NOT registered (WdspEngine hybrid owns RX dispatch until P4 — the B.6.b clobber class).	[DONE] Operator HL2 bench PASSED 2026-06-12.

Step	Status	What	Commit / gate
P4	[DONE]	Wire-LIVE switchover — **COMPLETE + operator HL2 bench PASSED 2026-06-13.** P4.a prep (2 wire-inert commits): sendProtocol1Samples verbatim in src/wire/NetworkProto1.cpp + io_keep_running + FrameComposer tail flipped to ReleaseSemaphore(hobbuffsRun) + SetTXAPostGen wdspcalls + TxReadBufp double* / hWriteThreadMain HANDLE. **P4.b SHIPPED** in 3 commits: `fb9ec41` Wire-LIVE RX-out gate, `0b551b4` first modulated RF through the direct-port chain, `9db2c50` SSB sideband fix (USB transmits USB, 9100-verified). Plus `84dbb12` TUN panadapter-spike fix (Thetis-exact zero-beat confirmed, display-only bug). **§7 retirements DONE** (`fe6a438` OutboundRing+Ep2SendThread, `99fd24a` txWorkerLoop+composeCC+keepalive, `e711256` old HL2-jack EP2 ring). **TCI digital TX re-home** `bd61d07` (TciTxBridge fills the §10.3 skeleton; on-air FT8 QSOs). Released v0.2.3.	Bench gates PASSED: SSB voice both sidebands + RF-safety kill-test + MSHV→TCI→FT8 on-air.

PureSignal is a committed feature — the P0.d `_cmaster` struct carries the full PS surface so v0.3 lands on top without a retrofit.

Historical stage index (pre-P0 lettered stages — superseded)

The original lettered plan below is retained for history. The P0 arc absorbed/superseded it: Stage C (ilv) = P0.b; Stage D (xcmaster) = P0.d; Stage E (TxChannel audit) = obsoleted by the P0.d TxChannel deletion; Stage B's idiom-translated AAMix was re-ported verbatim at P0.c.

Stage	Status	What	Notes
A	[DONE] DONE	CMaster.cpp wire-up stubs (Send*Outbound* + SetTCIRun)	Superseded by the P0.d full verbatim CMaster port.
B	[DONE] DONE	aamix.c direct port + wire into RX path	Shipped 2026-06-08 (idiom-translated); re-ported VERBATIM at P0.c.
C	[DONE] DONE (as P0.b)	ilv.c direct port (TX I/Q interleaver)	Landed verbatim at `wire/ILV.{h,cpp}`.
D	[DONE] DONE (as P0.d)	xcmaster pump body direct port	Landed verbatim in `wire/CMaster.cpp` (real stype/txid/chid + update[] CS).
E	[N/A] OBSOLETE	TxChannel reference-parity audit + reconciliation	TxChannel DELETED at P0.d — the verbatim create_xmtr replaces it.
F	[PEND] TBD	WDSP TXA chain components per CLAUDE.md §4.1 v0.2.1	compress / cfcomp / eqp / wcpagc — operator-driven per-feature direct port vs cffi-only call. Order TBD.

Stage	Status	What	Notes
G	[PEND] TBD	PureSignal direct ports (v0.3)	iqc / calcc / xbuilder / delay. Lands when v0.3 PS work starts. Needs operator HL2+ with the PS hardware mod.
H	[PEND] TBD	TX/PS auto-attenuator FSM port	C# study only (PSForm.cs); Lyra-native rewrite.

Ordering note: the P0->P4 sequence above is the active linear plan. F–H land after P4 Wire-LIVE. The methodology (smallest revertable step + mechanical diff + operator bench gate) applies within each step.

OPEN QUESTIONS FOR OPERATOR

These are decisions that shape the plan but I cannot make unilaterally — surface them when picking up tomorrow:

1. **Stage C (ilv.c) vs Stage D (xcmaster) vs Stage E

(TxChannel audit) — which first?** All three are defensible:

- Stage C unlocks reference-faithful TX I/Q flow.
- Stage D removes the Lyra-native pump (cleaner end-state).
- Stage E formalizes what's already shipped (lowest risk,

highest confidence-gain).

2. **Do we re-port networkproto1.c + network.c + the Step 14

wire pieces direct from reference, OR leave them as the current mix of Lyra-native rewrites + original `hl2_stream.cpp`?** The Step 14 plan was midway through these when Stage B started. Direct-port would replace the rewrites with reference-verbatim ports. Operator call.

3. **WDSP TXA-chain components — direct port (~150-600 LOC

each) vs cffi-only?** The DLL exports work; cffi is faster to ship. Direct port gives Lyra-cpp full source visibility + lets us patch behavior at the source level. Per-component operator decision; default = cffi-only unless a real source- level need surfaces.

4. **PS work timing — start v0.3 (iqc/calcc/etc.) now in

parallel with TX, or sequence after TX is fully done?** The CLAUDE.md plan defers PS to v0.3 (after v0.2 TX release). Operator may want to start the ports early.

5. **The §15.27 "opt-in toggles that become defaults" cleanup

from the Python project — does it have a lyra-cpp analog here?** The Python project had a long backlog of legacy toggles to retire. Lyra-cpp doesn't carry the same legacy, but new direct-port commits may obsolete some Settings surfaces operator-knows-about — track per-component.

SHIPPED-WORK LOG

Newest at top. Cross-references SESSION_LOG.md for full arc details.

2026-06-14 -> 2026-06-17 — post-P4: VAC v0.3.0/v0.3.1, native TX DSP rack, Profiles, TX monitor (#90), Out picker

These are post-direct-port work: native (Lyra-side) TX-DSP / host-audio / profile features that land ON TOP of the verbatim P0–P4 chain. They are NOT direct ports (no Thetis ChannelMaster/ WDSP source counterpart) — logged here as a post-P4 section so the P0–P4 port-stage record above stays intact.

• **VAC / IVAC (#102/#158)** — PC/VAC mic-input + RX-to-PC path.

`ivac.c` -> `wire/Ivac` direct port (Stage 0/1 `4831db7/52adff3`),

#158 complete (RX decode + TX on-air + auto-start + profile VAC source) `6c3fecb`. Released **v0.2.4** "Profiles + VAC" `c48c94d`; VAC mic mono-combine fix `b57bd09` -> **v0.2.5** `f55da2c`. Device-layer rework DL-0..DL-5 (PortAudio 19.7.0 vendored static, single full-duplex stream, Thetis-faithful host-API/device pickers, RX-muted-on-TX via SetIVACmox, old Qt two-stream `ivac_audio` retired): `11812f9`..`e8d0239`; plus two crash fixes — VAC-teardown UAF `f0c6497`, TCI dangling-owner `d508b8e`. TX MIC/ALC/LVL meters re-homed onto wire-live `GetTXAMeter` `993e10e`. Released **v0.3.0** `accee55`. Then **v0.3.1** `a3a517f`: VAC enable/disable crash fix + Vol/Mute routed to VAC `77dc55d` (#161) + Thetis-format TX MIC/ALC/LEV metering + #49 profile Leveler/ALC fields `9594bd5` (#160). v0.3.1 addresses the v0.3.0-banner "TX metering UNVERIFIED" loose-end in code (#160 bench close-out / Brent+Timmy field-confirm still in progress). Interim VAC-UI fixes (`88bdab1` relabel, `b57bd09` combine, `a01d308` output-"(none)") shipped along the way.

• **Stage F — native TX DSP rack (pre-WDSP-TXA mic rack):**

- **EQ #50** — native ParamEq (RBJ biquad cascade) `edbaa52`;

EqModel + EESDR3-style panel `17a28b7`; wired into the live TX mic rack `eb92231`.

- **Digital-mode auto-bypass** of the native mic DSP (DIGU/DIGL)

`c8441a6`.

- **Speech #88** — native SpeechProcessor (Auto-AGC leveller +

de-esser) `902b814`; Speech panel + EQ analyzer + Reset-All `93adb39`.

- **Combinator #51** — native engine `013bf12`; model+panel+dock

`61e16c4`; wired into the live rack after EQ `b53ae35`.

- **Plate #52** — native Schroeder–Moorer engine `ae9c5d3`;

PlateModel + "Plating" panel `0cfd7e5`; wired after Combinator `e96a6e8`.

- Rack-lamp dim in digital/CW + EQ analyzer recolor (#163)

`9be1fc1` / `558d5d9`.

• **Profiles #49 (TX/RX profile manager):** rack

saveState/loadState round-trip `e0f264e` (Stage 1); schema v3 bundling the native rack blobs `b453320` (Stage 2); rack wired into profile capture/apply `29f9b92` (Stage 3). Sideband- and band-agnostic SSB profile `296cafc`. Panel-lock float-drag fix `a82eb77` (exclude TX rack panels from LOCK) + lock also blocks resize/custom-float `cf976c1`. TX Speech profile capture + toggle-lamp refresh on recall (#162) `c17960c` / `46a428f`.

• **TX audio monitor #90** — Route 1 (MON button + Monitor slider,

HL2 jack) `6e6a282`; Routes 2+3 (VAC device + TCI RX-audio) `e1fd186`. Reference-verified against Thetis audio.cs.

• **#164** — inline Audio-panel "Out" output-device picker wired +

MON->MON-TX relabel `740f769`.

• USER_GUIDE catch-up commits (minor): `66adc5e`, `266d8cd`,

`dc45485`.

2026-06-13 — P4 Wire-LIVE switchover + v0.2.3 release ([DONE] HL2 bench PASSED)

• **P4.a prep** (wire-inert): sendProtocol1Samples verbatim

`a4aa8c3` + SetTXAPostGen/TxReadBufp/hWriteThreadMain field types `baef866`.

• **P4.b SHIPPED** (3 commits, operator HL2 bench PASSED 2026-06-13):

`fb9ec41` Wire-LIVE RX-out gate, `0b551b4` first modulated RF through the direct-port chain, `9db2c50` SSB sideband fix (USB transmits USB, 9100-verified). Plus `84dbb12` TUN panadapter-spike fix (display-only, Thetis-exact zero-beat).

• **§7 retirements DONE:** `fe6a438` OutboundRing+Ep2SendThread,

`99fd24a` txWorkerLoop+composeCC+keepalive, `e711256` old HL2-jack EP2 ring.

• **TCI digital TX re-home** `bd61d07` (first FT8 QSOs via MSHV/TCI

through the verbatim chain) + space-bar PTT toggle `2824ebf` (#157).

• Released **v0.2.3** "Transmit (SSB + digital TCI) on air" `e8aafbc` (tags `v0.2.3` + `tx-working-2026-06-13`).

2026-06-12 — P0.d cmbuffs/cmaster/cmsetup VERBATIM port ([DONE] HL2 bench PASSED)

• **P0.d** `afc7950` on `tx-rebuild`: NEW `wire/cmsetup.{h,cpp}`

(cmMAX* 16/4/4/2/32 + the 8 id helpers + SetRadioStructure + set_cmdefault_rates); `wire/CmBuffs.{h,cpp}` rewritten verbatim (`cmb`, `*CMB`, malloc0/_aligned_free; Phase-C retrofit deviations removed); `wire/CMaster.{h,cpp}` rewritten verbatim (FULL `_cmaster` struct incl. the PureSignal surface, `enum AudioCODEC`, raw TCI fn ptrs, verbatim create/destroy_cmaster + create/destroy_xmtr + xcmaster + all Sendp*/Set* setters).

• **TxChannel RAII carve-out DELETED** (`src/wdsp/TxChannel.{h,cpp}`

retired) — the verbatim create_xmtr opens the WDSP TXA channel + TX analyzer (disp 1) + out[0..2] + ILV itself through the wdspcalls seam.

• main.cpp: SetRadioStructure(2,1,1,1,0,…) config before

create_cmaster (chid(0,0)=0 RX1 / chid(1,0)=1 TX = the live layout); create_xmtr() post-resolve_wdsp_calls (documented DEFERRED-CALLSITE accommodation); handler-1.5 = gated destroy_xmtr().

• Verification: clean build ZERO warnings; test_ilv ALL PASS;

mechanical diffs IDENTICAL (CmBuffs.{h,cpp} / cmsetup.h / CMaster.h / all 10 fully-verbatim cmaster.c functions) + live-line-subset PASS for the partially-deferred bodies (sole accommodation: `(char*)""` at XCreateAnalyzer).

• New at startup: TWO cm_main pump threads (streams 0+1; stream 0

idles — no Inbound producer); TX stays wire-quiescent (OutboundTx null until P3).

• [DONE] **GATE PASSED 2026-06-12: operator HL2 RX-regression bench**

("RX still working") — P1 (obbuffs.c) is unblocked.

2026-06-09 -> 2026-06-11 — P0.a/P0.b/P0.c verbatim-rewrite arc

• Operator rejected the 2026-06-09 "rule-#8" retrofit deviations

("I said PORT!") -> every ported file rewritten VERBATIM.

• **P0.a** `e0c0f4a`: `wire/wdspcalls.{h,cpp}` — the single

approved WDSP linkage seam (fn-ptr table, exact export names, PureSignal entry family complete + dumpbin-verified). `scratch/_typedef_fidelity_test.cpp` disproved the claimed "C++ collisions" (twin typedef / `complex[2]` / raw fn ptr all compile verbatim).

• **P0.b** `1f49d98`: ilv.{h,c} verbatim -> `wire/ILV.{h,cpp}`;

new `wire/cmcomm.{h,cpp}` (complex/PORT/malloc0/PI); pilv[] bank gone; SendpOutboundTx raw fn ptr (register AFTER create_xmtr — no null guards, reference ordering). Mechanical diff whitespace-only; test_ilv ALL PASS.

• **P0.c** `e0f2584`: aamix.{h,c} verbatim -> `wire/AAMix.{h,cpp}`

(supersedes the Stage-B idiom translation); `wire/resample.h` = public RESAMPLE ABI; wdspcalls retyped + flush_resample; WdspEngine outbound lambda -> static fn + TU-scope context ptr. [DONE] **Operator HL2 RX bench PASSED 2026-06-11.**

2026-06-08 EOD — Stage B aamix.c port COMPLETE

• **Stage B (aamix.c)** [DONE] direct port shipped + bench-validated

• Commits `27ac2fa` -> `7a50b07` on `tx-rebuild`

• Operator HL2+ bench 19:27:49 -> 19:28:47: audible RX through

both AK4951 jack and PC-Soundcard sinks; mute/unmute via dispatch; output-device swap; AGC mode changes; chain out=N tracks 1:1 with dispatch calls=N across both sinks

• B.6.b root cause: Stage-A SendpOutboundRx no-op stub at

`RadioNet.cpp:272` clobbered `aaMix_->Outbound` ~1 ms after `WdspEngine::openRx1` wired the real lambda. Fixed by deleting the stub (commit `8b8e0da`).

• 2-stage diagnostic instrumentation (chain @ `225518c` +

dispatch @ `12369bc`) localized the defect to a single line of code on the 2nd bench — pure-speculation cost avoided.

• Methodology lessons locked into this doc § "LOCKED

METHODOLOGY" above (counters-first when symptom is "nothing happens"; operator-empirical rule held; "do as Thetis does" continues to pay).

• Tasks #130 (parent) + #132 (Stage B.6.b) closed.

• Backup `_backups/lyra-cpp-2026-06-08-aamix-stage-B-COMPLETE-with-docs.bundle`.

Earlier work

See `SESSION_LOG.md` for the full session-by-session log. The Step 14 wire-layer arc (Stages 1, 1.5, 2a, 2b1, 2b2) is covered in detail there + in `EXECUTION_PLAN.md`'s progress- tracking sections. TX-1 components 2-6 (`main` branch) are covered in `project_lyra_cpp_tx.md` (memory file).

RESUME POINTER (next session)

First actions (locked discipline):

1. `git fetch origin`

2. `git status` + `git log --oneline -5 tx-rebuild origin/main`

3. Read this doc + the "READ FIRST" block in

`project_lyra_cpp_tx.md`

4. Read `SESSION_LOG.md` newest entry for arc detail

Then — current open queue (one task): an **AM/DSB sideband bug** — AM and DSB modes produce only the upper sideband (lower sideband missing). RX-DSP investigation: the WDSP passband for AM/DSB is likely being set one-sided instead of symmetric (-W..+W around DC). Grounded design + reference- read + smallest-revertable-step + bench-gate, matching the Stage B arc.

Also pending operator discussion: **GitHub / main-branch posture** (`main` vs `tx-rebuild` reconciliation; current HEAD == `tx-rebuild` == `dc45485`).

The earlier post-P4 ordering (TX metering -> Stage F DSP rack -> Profiles -> RX2/Split -> PureSignal) is largely SHIPPED through v0.3.1: TX metering closed (#160), Stage F native DSP rack + Profiles #49 + TX monitor #90 all landed (see the SHIPPED-WORK LOG). Still PEND: RX2 + Split (#96–#101) and PureSignal (Stage G/H, v0.3) — both remain as the next major arcs after the AM/DSB fix.

Last updated: 2026-06-17 — post-P4 native work SHIPPED: VAC closeout v0.3.1 (TX metering #160 addresses the v0.3.0 UNVERIFIED loose-end in code; #160 bench close-out + tester field-confirm still in progress), Stage-F native TX DSP rack (EQ #50 / Speech #88 / Combinator #51 / Plate #52), Profiles #49, TX monitor #90 + Out picker #164. P1/obbuffs flipped [WIP]->[DONE] (wire-LIVE since P4.b). HEAD == tx-rebuild == `dc45485`. Active task: AM/DSB-only-upper-sideband RX-DSP bug. RX2/Split + PureSignal remain PEND.